

1 Solution — GIAM 3.6

Problem 1

1.1 Problem

Show that there is a perfect square that is the sum of two perfect squares.

1.2 Answer

$$25 = 9 + 16, \quad \text{that is} \quad 5^2 = 3^2 + 4^2.$$

1.3 Proof

The claim is existential,

$$\exists a, b, c \in \mathbb{Z}^+, \quad a^2 + b^2 = c^2,$$

so it is established by producing one triple and verifying it.

Proof. Take $a = 3$, $b = 4$, $c = 5$. These are positive integers, and

$$a^2 + b^2 = 9 + 16 = 25 = 5^2 = c^2.$$

So 25 is a perfect square that is the sum of the two perfect squares 9 and 16. ■

Nothing further is required. An existential statement needs a witness, not an argument.

1.4 The Picture

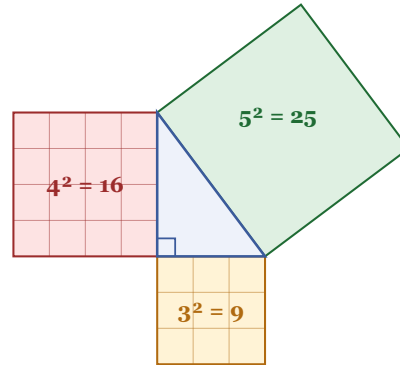
The classical illustration is to erect a square on each side of a right triangle with legs 3 and 4.

AN EXISTENTIAL CLAIM

some perfect square is the sum of two perfect squares

To prove $\exists$, exhibit one instance and check it. No general argument is needed, and none is expected.

$$3^2 + 4^2 = 9 + 16 = 25 = 5^2$$



legs 3 and 4, hypotenuse 5: the two small squares hold 9 and 16 unit cells — together exactly the 25 of the large one

ONE WITNESS IS ENOUGH — BUT THERE ARE MANY

Searching every c up to 100 finds 52 such triples, 16 of them primitive:

(3,4,5) (5,12,13) (8,15,17) (7,24,25) (20,21,29) (12,35,37) (9,40,41) (11,60,61) ...

Euclid's formula generates all of them: $a = m^2 - n^2$, $b = 2mn$, $c = m^2 + n^2$ for coprime $m > n$ of opposite parity. Taking $m = 2$, $n = 1$ returns (3, 4, 5); $m = 3$, $n = 2$ returns (5, 12, 13). Verified against the list.

THE SAME QUESTION FOR HIGHER POWERS HAS THE OPPOSITE ANSWER

Is some cube the sum of two cubes? Some fourth power the sum of two fourth powers? Searching every c below 400 for $k = 3, 4$ and 5 turns up nothing at all.

$$a^k + b^k = c^k \text{ has no positive solutions for any } k \geq 3$$

That is Fermat's Last Theorem — conjectured 1637, proved by Wiles in 1995. So $k = 2$ is the only exponent for which this exercise has an answer.

Figure 1.1: A diagram in three parts. The first states the existential claim — some perfect square is the sum of two perfect squares — and notes that to prove an existential statement one exhibits a single instance and checks it, with no general argument needed. The instance is 3 squared plus 4 squared equals 9 plus 16 equals 25 equals 5 squared. The centre is the classical picture: a right triangle with a vertical leg of length 4 and a horizontal leg of length 3, the right angle marked, and a square erected on each of the three sides. The square on the leg of length 4 is red and ruled into sixteen unit cells, labelled 4 squared equals 16; the square on the leg of length 3 is amber and ruled into nine unit cells, labelled 3 squared equals 9; the square on the hypotenuse of length 5 is green, labelled 5 squared equals 25. The caption notes that the two small squares hold 16 and 9 unit cells, together exactly the 25 of the large one. A band below observes that one witness is enough but there are many: searching every hypotenuse up to 100 finds 52 such triples, 16 of them primitive, beginning 3 4 5; 5 12 13; 8 15 17; 7 24 25; 20 21 29; 12 35 37; 9 40 41; and 11 60 61. Euclid's formula generates all of them, with a equal to m squared minus n squared, b equal to $2mn$, and c equal to m squared plus n squared for coprime m greater than n of opposite parity; taking m equals 2 and n equals 1 returns 3, 4, 5, and m equals 3 with n equals 2 returns 5, 12,

13, verified against the list. A final band notes that the same question for higher powers has the opposite answer: searching every c below 400 for exponents 3, 4 and 5 turns up nothing at all, because a to the k plus b to the k equals c to the k has no positive solutions for any k at least 3. That is Fermat's Last Theorem, conjectured in 1637 and proved by Wiles in 1995, so k equals 2 is the only exponent for which this exercise has an answer.

1.5 Remark — What an Existential Proof Costs

This is the cheapest kind of proof there is, and it is worth being explicit about why. The statement quantifies existentially:

$$\exists a, b, c, \ a^2 + b^2 = c^2.$$

To establish it one must supply an object and verify the property. Both steps are finite and mechanical — here, one addition and one comparison. Contrast the two directions:

statement	to prove	to refute
$\exists x, P(x)$	exhibit one x	a general argument
$\forall x, P(x)$	a general argument	exhibit one x

So the work in an existential problem lies entirely in *finding* the witness, never in justifying it. Here the witness is famous enough to be recalled rather than searched for, but even a blind search over $a, b, c \leq 5$ would locate it in a few dozen trials.

Note also what the problem does **not** ask. It does not ask how many such squares there are, nor for a characterisation of them, nor whether the analogous statement holds for cubes. Each of those is a strictly harder question, and the last one took three and a half centuries.

1.6 Remark — Witnesses Are Abundant

The single triple $(3, 4, 5)$ discharges the problem, but it is far from isolated. Searching every hypotenuse $c \leq 100$ turns up **52** Pythagorean triples, of which **16** are primitive (no common factor):

$$(3, 4, 5), \ (5, 12, 13), \ (8, 15, 17), \ (7, 24, 25), \\ (20, 21, 29), \ (12, 35, 37), \ (9, 40, 41), \ (11, 60, 61), \ \dots$$

The remaining **36** are multiples of these, such as $(6, 8, 10)$ and $(9, 12, 15)$ — each supplying another perfect square that is a sum of two.

Every primitive triple arises from **Euclid's parametrisation**: for coprime integers $m > n > 0$ of opposite parity,

$$a = m^2 - n^2, \quad b = 2mn, \quad c = m^2 + n^2,$$

which satisfies $a^2 + b^2 = (m^2 + n^2)^2 = c^2$ identically. Running m up to **11** reproduces exactly the sixteen primitive triples found by brute search. Taking $m = 2$, $n = 1$ gives the witness used above.

Since there are infinitely many admissible (m, n) , there are infinitely many such squares — a much stronger statement than the one asked for, and available almost for free once the parametrisation is known.

1.7 Remark — Why the Even Leg Is Always There

The formula makes visible something proved by congruences in 3.4 Problem 8: in any Pythagorean triple, at least one of a, b is even. Here the leg $b = 2mn$ carries a factor of **2** by construction, and since m and n have opposite parity one of them is even, so in fact $4 \mid b$.

The two arguments answer different questions. The mod-4 proof of 3.4 Problem 8 shows an all-odd triple is *impossible*, using nothing but the equation. Euclid's formula shows *where the even leg comes from*, but presupposes the classification. For the witness $(3, 4, 5)$: $m = 2$, $n = 1$, and indeed $b = 4$ is divisible by 4.

1.8 Remark — The Same Question for Cubes

The natural sequel is to ask whether some cube is the sum of two cubes, or some fourth power the sum of two fourth powers. Searching every $c < 400$ for $k = 3, 4, 5$ finds **nothing**, and this is not an artefact of the range:

$$a^k + b^k = c^k \quad \text{has no solution in positive integers for any } k \geq 3.$$

That is **Fermat's Last Theorem** — asserted in a margin in 1637 and finally proved by Andrew Wiles in 1995. So the exponent **2** is the *only* one for which this exercise has an answer at all, and the contrast is stark:

k	solutions	status
2	infinitely many	Euclid, $\sim$ 300 BCE
≥ 3	none	Wiles, 1995

There is a lesson in the pairing. Proving the $k = 2$ case took one line because the answer was *yes* and a witness existed. Proving the $k \geq 3$ case took centuries because the answer was *no*, and a negative existential is a universal statement in disguise — one must rule out every triple, not exhibit one. The quantifier asymmetry that made this problem trivial is exactly what made Fermat's version hard.

Near misses do exist and are worth seeing, since they show the failure is not obvious: $6^3 + 8^3 = 728$, one short of $9^3 = 729$.